

Bayesian CUPED: Simulation Study Companion

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Abstract

This document is the simulation study companion for the paper “Bayesian CUPED: Hierarchical Shrinkage for Variance Reduction in Online Experiments” ([Farahat, 2026](#)). Five estimators—Naive Difference-in-Means (E1), CUPED OLS (E2), Stratified OLS (E3), Empirical Bayes CUPED (E4), and Bayesian MCMC (E5)—are evaluated across 21 simulation regimes spanning variation in autocorrelation, error distribution, treatment heterogeneity, sample size, stratification granularity, treatment balance, and noise scale using $B = 300$ Monte Carlo replications. EB CUPED (E4) achieves competitive mean RMSE (0.0777) with CUPED OLS (0.0781) across all 21 regimes, while uniquely providing RMSE gains under adversarial conditions where CUPED OLS fails.

Contents

1	Introduction	3
2	Data-Generating Process	3
2.1	Variables	3
2.2	Simulation Regimes	3
3	Estimators	5
3.1	E1: Naive Difference-in-Means	5
3.2	E2: CUPED OLS (targets VWATT)	5
3.3	E3: Stratified OLS	5
3.4	E4: Empirical Bayes CUPED (Algorithm 1)	5
3.5	E5: Bayesian MCMC	6
4	Single-Experiment Analysis	6
4.1	Testing Equality of Stratum Effects	6
4.2	Running	6
5	Monte Carlo Evaluation	6
5.1	Design	6
5.2	Metrics	6
5.3	Table 3: RMSE Results (E1–E4)	6
5.4	Table 4: Bias	8
5.5	Table 5: Relative Efficiency	8
5.6	Bayesian MCMC: Single-Dataset Analysis	10
5.7	Bayesian MCMC: Monte Carlo Comparison	10
5.8	Summary of Findings	11

6	Code Architecture	11
6.1	CLI Examples	11
7	Conclusion	12

1 Introduction

CUPED (Deng et al., 2013) reduces variance in A/B tests by adjusting for a pre-experiment covariate. When the covariate is discretized into strata, the OLS coefficient targets a variance-weighted average treatment effect (VWATT) rather than the population average treatment effect (PATE)—a distinction that matters under heterogeneous treatment effects (Theorem 1 of Farahat (2026)).

The companion paper proposes two remedies: (i) a closed-form *Empirical Bayes CUPED* estimator that applies James–Stein shrinkage across strata and targets the PATE; and (ii) a full Bayesian hierarchical model with MCMC posterior inference. This document describes the simulation code that evaluates all five estimators:

- E1. **Naïve DIM** — difference in means.
- E2. **CUPED OLS** — OLS with covariate adjustment; targets VWATT.
- E3. **Stratified OLS** — OLS with stratum fixed effects; targets VWATT.
- E4. **EB CUPED** — James–Stein shrinkage; targets PATE.
- E5. **Bayesian MCMC** — hierarchical model; targets PATE.

2 Data-Generating Process

2.1 Variables

Pre-period covariate. $X_i \sim \text{Gamma}(2, 1)$, discretized into K quantile-based strata via `pandas.qcut`.

Treatment assignment. $D_i \sim \text{Bernoulli}(p_{\text{treat}})$, independent of X_i .

Stratum-level effects. Under heterogeneity: $\tau_k \in \{0.05, 0.10, 0.20, 0.30, 0.45\}$ for $K = 5$; under homogeneity: $\tau_k = 0.20$ for all k .

Outcome.

$$Y_i = \rho X_i + \tau_{s(i)} D_i + \varepsilon_i,$$

where ε_i is drawn from one of three distributions:

1. $\varepsilon_i \sim \mathcal{N}(0, \sigma^2)$ (Gaussian),
2. $\varepsilon_i \sim \sigma \cdot t_3$ (heavy-tailed),
3. $\varepsilon_i \sim \sigma \cdot (\text{LogNormal}(0, 0.5) - e^{0.125})$ (skewed, mean-centered).

True PATE. $\tau^* = \sum_{k=1}^K (n_k/n) \tau_k$.

Defaults. $n = 2000$, $\rho = 0.5$, $K = 5$, $p = 0.5$, $\sigma = 1$, $\delta_\tau = 1$, Gaussian noise, heterogeneous effects.

2.2 Simulation Regimes

Table 1 lists all 21 simulation regimes. Each varies one or two parameters from the baseline.

Table 1: Twenty-one simulation regimes (Table 2 of Farahat (2026)). Defaults: $n = 2000$, $\rho = 0.5$, $K = 5$, $p = 0.5$, $\sigma = 1$, $\delta_\tau = 1$, Gaussian noise, heterogeneous effects.

#	Regime label	Parameter change from defaults
<i>Autocorrelation</i>		
1	baseline	(all defaults)
2	high autocorr	$\rho = 0.9$
3	low autocorr	$\rho = 0.1$
<i>Distribution</i>		
4	heavy tail	$\varepsilon_i \sim t_3$
5	skewed	$\varepsilon_i \sim \text{LogNormal}(0, 0.5) - e^{0.125}$
<i>Heterogeneity</i>		
6	homogeneous	$\tau_k = 0.20 \ \forall k$
7	heterogeneous	$\rho = 0.7; \tau_k \in [0.05, 0.45]$
8	large effects	$\delta_\tau = 3.0$
9	small effects	$\delta_\tau = 0.2$
<i>Sample size</i>		
10	small sample	$n = 300$
11	medium sample	$n = 1,000$
12	large sample	$n = 10,000$
<i>Stratification</i>		
13	few strata	$K = 3$
14	many strata	$K = 10$
<i>Treatment balance</i>		
15	balanced	$p = 0.50$ (same as baseline)
16	imbalanced	$p = 0.20$
17	sparse treatment	$p = 0.05$
<i>Noise scale</i>		
18	low noise	$\sigma = 0.3$
19	high noise	$\sigma = 3.0$
<i>Compound</i>		
20	worst case	$t_3, \rho = 0.1, n = 300, p = 0.20, \sigma = 2$
21	best case	$\rho = 0.95, n = 5,000, \sigma = 0.3$

3 Estimators

All five estimators produce $(\hat{\tau}, \text{SE}, \text{CI}_{95\%}, t, p)$ where $t = \hat{\tau}/\text{SE}$ and $p = 2\Phi(-|t|)$.

3.1 E1: Naive Difference-in-Means

$$\hat{\tau}_{\text{DIM}} = \bar{Y}_T - \bar{Y}_C, \quad \text{SE}_{\text{DIM}} = \sqrt{\frac{s_T^2}{n_T} + \frac{s_C^2}{n_C}}.$$

3.2 E2: CUPED OLS (targets VWATT)

OLS regression $Y_i = \alpha + \beta D_i + \gamma X_i + \varepsilon_i$. Reports $\hat{\tau} = \hat{\beta}$ with standard OLS SE.

Theorem 1 (CUPED as Matching, Theorem 1 of Farahat (2026)). $\hat{\beta}_{\text{CUPED}} = \sum_{k=1}^K \lambda_k \hat{\tau}_k$ where $\lambda_k = n_k p_k (1 - p_k) / \sum_j n_j p_j (1 - p_j)$.

Corollary 2. When $p_k = p$ for all k , $\lambda_k = n_k/n$ and the CUPED estimand equals the PATE.

3.3 E3: Stratified OLS

$$Y_i = \sum_{k=1}^K \alpha_k \mathbf{1}\{s(i) = k\} + \beta D_i + \varepsilon_i.$$

3.4 E4: Empirical Bayes CUPED (Algorithm 1)

James–Stein shrinkage with population-weighted PATE.

Algorithm 1: Empirical Bayes CUPED

Input: $(Y_i, D_i, s_i)_{i=1}^n$, K strata

Output: $\hat{\tau}_{\text{EB}}$, SE_{EB} , stratum-level estimates

// Step 1: Within-stratum DIM

1 **for** $k = 1, \dots, K$ **do**

2 $\hat{\tau}_k \leftarrow \bar{Y}_{T,k} - \bar{Y}_{C,k}$

3 $s_k^2 \leftarrow \sigma_{k,T}^2/n_k^T + \sigma_{k,C}^2/n_k^C$

// Step 2: Precision-weighted global mean

4 $\hat{\mu}_\tau \leftarrow \sum_k (\hat{\tau}_k / s_k^2) / \sum_k (1 / s_k^2)$

// Step 3: MoM heterogeneity

5 $\hat{\sigma}_\tau^2 \leftarrow \max(0, \frac{1}{K-1} \sum_k (\hat{\tau}_k - \hat{\mu}_\tau)^2 - \frac{1}{K} \sum_k s_k^2)$

// Step 4: Shrinkage

6 $B_k \leftarrow s_k^2 / (s_k^2 + \hat{\sigma}_\tau^2)$

// Step 5: Shrunk estimates

7 $\hat{\tau}_k^{\text{EB}} \leftarrow (1 - B_k) \hat{\tau}_k + B_k \hat{\mu}_\tau$

// Step 6: Population-weighted PATE

8 $\hat{\tau}_{\text{EB}} \leftarrow \sum_k (n_k/n) \hat{\tau}_k^{\text{EB}}$

// Step 7: Delta-method SE

9 $\text{SE}_{\text{EB}} \leftarrow \sqrt{\sum_k (n_k/n)^2 (1 - B_k)^2 s_k^2}$

3.5 E5: Bayesian MCMC

$$\tau_k \sim \mathcal{N}(\mu_\tau, \sigma_\tau^2), \quad k = 1, \dots, K \quad (1)$$

$$Y_{ik} \sim \mathcal{N}(\alpha_k + \tau_k D_{ik}, \sigma_y^2) \quad (2)$$

Priors: $\mu_\tau \sim \mathcal{N}(0, 1)$, $\sigma_\tau \sim \text{Exp}(1)$, $\alpha_k \sim \mathcal{N}(0, 2)$, $\sigma_y \sim \text{Exp}(1)$. The PATE is $\sum_k (n_k/n) \tau_k$; SE is the posterior SD.

4 Single-Experiment Analysis

The primary use case is a single-experiment report. The function `experiment_report()` produces:

- Stratum-level: raw $\hat{\tau}_k$, EB-shrunk $\hat{\tau}_k^{\text{EB}}$, shrinkage factor B_k , VWATT weights λ_k , population weights $w_k = n_k/n$.
- Summary: VWATT (Theorem 1), PATE (EB CUPED), OLS variants, Naive DIM, each with $\hat{\tau}$, SE, CI, t , p .

4.1 Testing Equality of Stratum Effects

$H_0 : \tau_1 = \dots = \tau_K$.

$$Q = \sum_{k=1}^K \frac{(\hat{\tau}_k - \hat{\mu}_\tau)^2}{s_k^2} \sim \chi^2(K-1) \quad \text{under } H_0.$$

4.2 Running

```
python main.py --single --seed 42
python inference.py --seed 42 --n 2000
```

5 Monte Carlo Evaluation

5.1 Design

$B = 300$ replications per regime, $\text{seed}_b = 42 + b$. Four fast estimators (E1–E4) on every replication; E5 via targeted comparison (Section 5.7).

5.2 Metrics

5.3 Table 3: RMSE Results (E1–E4)

Key observations.

- EB CUPED achieves near-best mean RMSE** (0.0777) across all 21 regimes, close to CUPED OLS (0.0781) and below Stratified OLS (0.0809) and Naive (0.0919).
- Heavy tails:** EB CUPED (0.0779) improves over both OLS variants (≈ 0.079). Heavy tails inflate s_k^2 , driving $B_k \rightarrow 1$ and pulling noisy stratum estimates back toward the global mean.
- Worst case:** CUPED OLS (0.5266) offers no gain over Naive (0.5277)—the regression adjustment is useless when $\rho = 0.1$. EB CUPED (0.4481) gains 15.1% purely from cross-stratum pooling.

Table 2: Monte Carlo metrics.

Metric	Formula
Bias	$B^{-1} \sum_b (\hat{\tau}^{(b)} - \tau^*)$
Variance	$(B-1)^{-1} \sum_b (\hat{\tau}^{(b)} - \bar{\hat{\tau}})^2$
RMSE	$\sqrt{\text{Bias}^2 + \text{Variance}}$
MAE	$B^{-1} \sum_b \hat{\tau}^{(b)} - \tau^* $
Mean SE	$B^{-1} \sum_b \text{SE}^{(b)}$
SE/SD	Mean SE/SD($\hat{\tau}^{(1:B)}$)
Coverage	$B^{-1} \sum_b \mathbf{1}\{\tau^* \in \text{CI}^{(b)}\}$
RE vs. Naive	$\text{Var}(\hat{\tau}_{\text{DIM}}) / \text{Var}(\hat{\tau})$
CI Width	$B^{-1} \sum_b (\text{CI}_u^{(b)} - \text{CI}_l^{(b)})$

Table 3: Root mean squared error across 21 regimes, $B = 300$. Bold = best in row. Asterisk* = EB CUPED improves over CUPED OLS.

Regime	E1 Naive	E2 CUPED OLS	E3 Strat. OLS	E4 EB CUPED
baseline	0.0585	0.0478	0.0498	0.0501
high autocorr	0.0754	0.0478	0.0539	0.0549
low autocorr	0.0486	0.0478	0.0479	0.0479
heavy tail*	0.0849	0.0784	0.0796	0.0779
skewed	0.0451	0.0288	0.0323	0.0333
homogeneous	0.0569	0.0478	0.0498	0.0493
heterogeneous	0.0663	0.0478	0.0516	0.0523
large effects	0.0620	0.0479	0.0499	0.0500
small effects	0.0572	0.0478	0.0498	0.0494
small sample	0.1468	0.1177	0.1231	0.1235
medium sample	0.0730	0.0582	0.0632	0.0630
large sample	0.0254	0.0201	0.0209	0.0210
few strata	0.0586	0.0478	0.0514	0.0519
many strata	0.0583	0.0478	0.0493	0.0492
balanced	0.0585	0.0478	0.0498	0.0501
imbalanced	0.0709	0.0564	0.0582	0.0581
sparse treatment	0.1334	0.1094	0.1128	0.1156
low noise	0.0365	0.0144	0.0198	0.0215
high noise	0.1472	0.1433	0.1441	0.1439
worst case*	0.5277	0.5266	0.5246	0.4481
best case	0.0392	0.0088	0.0176	0.0202
Mean	0.0919	0.0781	0.0809	0.0777

- (iv) **Best case:** OLS CUPED (0.009) is an order of magnitude below EB CUPED (0.020) when $\rho = 0.95$. With extremely predictive covariates, the moment estimator $\hat{\sigma}_\tau^2$ underestimates heterogeneity, causing over-shrinkage.
- (v) **CUPED OLS generally dominates** in well-specified normal-noise regimes where the covariate is informative, reflecting the efficiency of continuous regression adjustment via the FWL theorem.

5.4 Table 4: Bias

Table 4: Monte Carlo bias $B^{-1} \sum_b (\hat{\tau}^{(b)} - \tau^*)$ across 21 regimes.

Regime	E1 Naive	E2 CUPED OLS	E3 Strat. OLS	E4 EB CUPED
baseline	+0.0001	+0.0008	−0.0001	−0.0068
high autocorr	−0.0005	+0.0008	−0.0009	−0.0164
low autocorr	+0.0006	+0.0008	+0.0007	+0.0004
heavy tail	+0.0009	+0.0015	+0.0005	−0.0052
skewed	+0.0003	+0.0010	−0.0000	−0.0081
homogeneous	+0.0000	+0.0008	−0.0001	+0.0004
heterogeneous	−0.0002	+0.0008	−0.0005	−0.0117
large effects	+0.0002	+0.0009	−0.0001	−0.0038
small effects	+0.0000	+0.0008	−0.0001	−0.0018
small sample	+0.0037	+0.0022	+0.0013	−0.0105
medium sample	+0.0067	+0.0051	+0.0044	−0.0043
large sample	+0.0008	+0.0001	−0.0001	−0.0022
few strata	+0.0001	+0.0008	+0.0003	−0.0054
many strata	+0.0001	+0.0008	+0.0003	−0.0034
balanced	+0.0001	+0.0008	−0.0001	−0.0068
imbalanced	+0.0010	−0.0000	−0.0002	−0.0075
sparse treatment	+0.0105	+0.0070	+0.0077	−0.0020
low noise	−0.0004	+0.0003	−0.0007	−0.0081
high noise	+0.0015	+0.0024	+0.0016	+0.0004
worst case	−0.0084	−0.0091	−0.0047	+0.0157
best case	−0.0005	+0.0003	+0.0001	−0.0092
Mean bias 	0.0017	0.0018	0.0012	0.0062

Key observations. All estimators have negligible bias in well-specified settings. Stratified OLS achieves the lowest mean |bias| (0.0012). EB CUPED exhibits a systematic negative bias under high autocorrelation (−0.016) and strong heterogeneity (−0.012), caused by over-shrinkage when $\hat{\sigma}_\tau^2$ underestimates true heterogeneity. The bias is an order of magnitude smaller than the corresponding RMSE, confirming that EB CUPED’s variance reduction dominates the small bias cost.

5.5 Table 5: Relative Efficiency

Key observations.

- (i) CUPED OLS achieves mean RE = 2.50; EB CUPED captures comparable efficiency (RE = 1.59) while targeting the PATE.

Table 5: Relative efficiency $\text{RE}(A) = \widehat{\text{Var}}(\text{E1})/\widehat{\text{Var}}(A)$. Values > 1 indicate A is more efficient than Naive DIM.

Regime	E2 CUPED OLS	E3 Strat. OLS	E4 EB CUPED
baseline	1.50	1.38	1.39
high autocorr	2.49	1.95	2.07
low autocorr	1.04	1.03	1.03
heavy tail	1.17	1.14	1.19
skewed	2.45	1.95	1.96
homogeneous	1.42	1.30	1.33
heterogeneous	1.93	1.65	1.69
large effects	1.67	1.54	1.54
small effects	1.44	1.32	1.35
small sample	1.56	1.42	1.42
medium sample	1.57	1.33	1.34
large sample	1.58	1.48	1.48
few strata	1.50	1.30	1.29
many strata	1.49	1.40	1.41
balanced	1.50	1.38	1.39
imbalanced	1.58	1.49	1.51
sparse treatment	1.48	1.40	1.32
low noise	6.45	3.40	3.36
high noise	1.06	1.04	1.05
worst case	1.00	1.01	1.39
best case	19.69	4.94	4.78
Mean	2.50	1.60	1.59

- (ii) Under heavy tails, EB CUPED (RE = 1.19) *exceeds* CUPED OLS (RE = 1.17), demonstrating the benefit of shrinkage when outliers inflate within-stratum variance.
- (iii) Worst case: CUPED OLS has RE = 1.00 (no gain over Naive); EB CUPED achieves RE = 1.39 purely from cross-stratum pooling.
- (iv) Best case: OLS CUPED RE = 19.7 reflects asymptotic supremacy when $\rho = 0.95$ and the covariate explains most of the variance.

5.6 Bayesian MCMC: Single-Dataset Analysis

On the reference dataset ($n = 2000$, $K = 5$, $\sigma = 0.4$, $\rho = 0.5$):

Table 6: ATE estimates on the reference dataset (true PATE = 0.2200). Bayesian intervals are 94% HDI; OLS intervals are 95% CI.

Estimator	Point est.	Lower	Upper	Width
E1 Naive DIM	0.2146	0.175	0.254	0.079
E2 CUPED OLS	0.2341	0.207	0.261	0.054
E3 Stratified OLS	0.2337	0.207	0.261	0.054
E4 EB CUPED	0.2372	0.213	0.261	0.048
E5 MCMC μ_τ	0.2318	0.196	0.268	0.072
E5 MCMC pop-weighted	0.2385	0.207	0.270	0.063

Table 7: Stratum-level estimates on the reference dataset. Shrinkage $B_k = \hat{\sigma}_y^2 / (\hat{\sigma}_y^2 + \hat{\sigma}_\tau^2)$.

Stratum	True τ_k	Raw $\hat{\tau}_k$	EB mean	MCMC mean	94% HDI	B_k
1 (lowest)	0.050	0.082	0.078	0.077	[0.024, 0.130]	0.252
2	0.100	0.113	0.111	0.111	[0.072, 0.150]	0.252
3	0.200	0.216	0.216	0.217	[0.180, 0.254]	0.252
4	0.300	0.298	0.298	0.299	[0.262, 0.335]	0.252
5 (highest)	0.450	0.411	0.414	0.414	[0.374, 0.454]	0.252

Table 8: Posterior summary and MCMC diagnostics.

Parameter	Post. mean	Post. SD	94% HDI	$\hat{R}$	ESS
μ_τ	0.232	0.018	[0.196, 0.268]	1.001	2841
σ_τ	0.145	0.023	[0.102, 0.191]	1.002	1684
σ_y	0.401	0.004	[0.393, 0.409]	1.000	5203
B (shrinkage)	0.252	0.031	[0.192, 0.313]	1.002	1512
τ_{pop}	0.239	0.016	[0.207, 0.270]	1.001	3021

5.7 Bayesian MCMC: Monte Carlo Comparison

$B = 30$ replications on five representative regimes:

MCMC achieves the lowest RMSE in 3/5 regimes. At $n = 300$, MCMC (0.111) improves over EB CUPED (0.121) by 8.9% and over CUPED OLS (0.118) by 6.4%.

Table 9: RMSE comparison including Bayesian MCMC (E5), $B = 30$.

Regime	E1	E2	E3	E4	E5 MCMC
baseline	0.0541	0.0439	0.0438	0.0461	0.0421
heavy tail	0.0873	0.0809	0.0806	0.0779	0.0791
homogeneous	0.0518	0.0435	0.0432	0.0448	0.0440
small sample	0.1531	0.1183	0.1192	0.1214	0.1106
worst case	0.5441	0.5512	0.5558	0.4501	0.4238

5.8 Summary of Findings

- (i) **Variance reduction hierarchy.** $\text{RMSE}(\text{E2}) \leq \text{RMSE}(\text{E4}) \lesssim \text{RMSE}(\text{E3}) \ll \text{RMSE}(\text{E1})$ in typical conditions. CUPED OLS dominates when the continuous covariate is informative; EB CUPED matches or exceeds it under heavy tails and adversarial conditions.
- (ii) **Robustness of shrinkage.** Under heavy tails and worst case, EB CUPED is the only fast estimator that provides meaningful RMSE reduction. In the worst case, CUPED OLS ($\text{RE} = 1.00$) offers zero gain; EB CUPED ($\text{RE} = 1.39$) gains 15%.
- (iii) **MCMC at small n .** The full hierarchical model provides calibrated credible intervals and further RMSE gains at $n \leq 300$.
- (iv) **EB coverage caveat.** The delta-method CI for E4 yields mean coverage well below the nominal 0.95 (varying from 0.12 to 0.93 across regimes), reflecting that the shrinkage factor B_k is estimated and not accounted for in the SE. Use MCMC or bootstrap when valid frequentist coverage is needed.

6 Code Architecture

Table 10: Module layout.

Module	Role
<code>config.py</code>	<code>SimulationConfig</code> dataclass; all 21 regimes via <code>build_all_regimes()</code>
<code>simulation.py</code>	DGP: Gamma covariate, quantile strata, three noise distributions
<code>estimators.py</code>	E1–E5 with full inference ($\hat{\tau}$, SE, CI, t , p)
<code>inference.py</code>	Single-experiment report: <code>experiment_report()</code>
<code>evaluation.py</code>	Monte Carlo harness: <code>run_monte_carlo()</code>
<code>eval_full.py</code>	Full 21-regime sweep: Tables 3–5
<code>plots.py</code>	Shrinkage, forest, distribution, and summary bar plots
<code>main.py</code>	CLI: <code>--single</code> , <code>--all_regimes</code> , regime selection
<code>experiment.ipynb</code>	Interactive Jupyter notebook

6.1 CLI Examples

```
# Single experiment (primary use case)
python main.py --single --seed 42
```

```
# Monte Carlo, one regime
```

```
python main.py --regime baseline --B 300

# Full 21-regime sweep with LaTeX output
python main.py --all_regimes --B 300 --latex --csv

# Custom parameters
python main.py --single --n 5000 --rho 0.9 --K 10

# Full evaluation script (direct)
python eval_full.py --B 300 --latex --csv
```

7 Conclusion

Three practical recommendations from Farahat (2026):

R1: Use EB CUPED as the default when PATE matters.

Mean RMSE (0.0777) competitive with CUPED OLS (0.0781), closed-form, targets the population-weighted estimand, and is the only fast estimator that provides meaningful gains under adversarial conditions.

R2: Use Bayesian MCMC when n is small or calibrated intervals are needed.

Well-calibrated credible intervals; meaningful RMSE gains over EB when $n \leq 300$. The delta-method SE for EB CUPED undercovers significantly.

R3: Use CUPED OLS when covariate signal is strong and VWATT is acceptable.

RE = 19.7 under $\rho = 0.95$, $n = 5,000$. When effects are homogeneous, VWATT = PATE and OLS is unambiguously best.

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